

Juan David Munoz Bolanos

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

March 19, 2026

ASML

Hiring Manager

De Run 6501

5504 Veldhoven

Netherlands

Subject: Application for Optical Systems Engineer

Dear Hiring Manager,

As an optical engineer with a Ph.D. specializing in wavefront sensing, interferometric metrology, and opto-mechatronic system integration, I am eager to bring my expertise to ASML. I am drawn to ASML's mission of pushing the physical limits of precision optical systems, and I am confident that my hands-on experience in diffraction-limited metrology and system integration aligns directly with the challenges your teams face in EUV lithography development.

During my Ph.D. at the Medical University of Innsbruck, I designed and built from scratch a fully automated wavefront sensing and phase-retrieval system — writing the complete C++/Python/LabVIEW control stack, integrating 63-actuator Deformable Phase Plates and spatial light modulators, and implementing iterative interferometric (F-SHARP) algorithms to achieve diffraction-limited resolution. Optical performance was validated through custom Python wave-field simulations based on Zernike decomposition, supported by familiarity with ZEMAX OpticStudio for tolerancing. This work resulted in two peer-reviewed publications (ACS Photonics 2025, Biomedical Optics Express 2023) and directly mirrors the kind of precision optical sensing and aberration-correction challenges tackled in EUV lithography systems.

Prior to my Ph.D., I prototyped a dispersive 1064 nm fiber probe Raman spectrometer at IPHT Jena — handling optical layout through detector calibration — and deployed Halcon-based machine vision inspection pipelines in a high-speed industrial setting at INTECOL S.A.S. This combination of academic photonics depth, hands-on metrology, and industrial systems thinking positions me well to contribute to ASML's optical module development and qualification workflows.

I am completing my Ph.D. in May 2026 and am available to relocate to Veldhoven. I would welcome the opportunity to discuss how my background in high-precision optical metrology and system integration can support your teams.

Sincerely,

Juan David Munoz Bolanos

Optical Systems Engineer

Medical University of Innsbruck

Juan David Munoz Bolanos

Optical Systems Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | LinkedIn Profile



PROFESSIONAL SUMMARY

Optical Systems Engineer with a Ph.D. in adaptive optics and a track record of end-to-end precision instrument development: from interferometric wavefront sensing through C++/Python/LabVIEW integration and peer-reviewed validation. Deep expertise in optical wave-field simulation and Zernike-based aberration modeling; familiar with ZEMAX OpticStudio for tolerancing. ZEISS Summer School alumnus. Available May 2026.

TECHNICAL COMPETENCIES

Wavefront Sensing & Interferometric Metrology

Designed and deployed F-SHARP interferometry and batched-DASH wavefront optimization for sub-wavelength aberration correction. Novel spatial-shift algorithm ("3N method") enabling high-accuracy phase retrieval in diffraction-limited systems.

Optical Simulation & Modeling

Deep knowledge of wave-field propagation modeling and Zernike polynomial decomposition, implemented in custom Python simulation frameworks. Familiar with ZEMAX OpticStudio for optical tolerancing and aberration sensitivity analysis.

Opto-Mechatronic Integration & Automation

Full hardware-software integration of adaptive optics systems (deformable mirrors, SLMs, galvo scanners, scientific cameras) via C++/Python/LabVIEW. Automated calibration and alignment routines ensuring repeatable, high-precision performance.

Machine Vision & Industrial Inspection

Industrial deployment of Halcon and OpenCV pipelines for high-speed automated inspection and defect detection. Custom illumination design (bright-field, dark-field, ring-light) for challenging surface-inspection tasks.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Optical Systems Engineer (PhD Researcher)

Innsbruck, Austria

Jan 2022 – Present

- **Wavefront Sensing System (ACS Photonics '25 / Biomed. Opt. Expr. '23):** Designed and built from scratch a fully automated interferometric wavefront sensing platform. Integrated 63-actuator Deformable Phase Plates and TI Phase Light Modulators via a custom C++/Python/LabVIEW stack. Achieved diffraction-limited resolution in multi-photon deep-tissue imaging, validated through two peer-reviewed publications.
- **Interferometry & Phase Retrieval:** Implemented F-SHARP wavefront interferometry and developed a novel spatial-shift estimation algorithm ("3N method") enabling high-fidelity Zernike decomposition of experimental aberrations. Benchmarked against custom Python wave-propagation simulations.
- **Optical Simulation & Tolerancing:** Built custom Python wave-field simulation frameworks using Zernike decomposition to predict and characterise aberration sensitivity throughout the optical path. Familiar with ZEMAX OpticStudio for optical tolerancing support.
- **System Procurement & Qualification:** Specified and procured high-precision opto-mechanical components, DAQ electronics, and interferometric sensors. Validated incoming components against technical specifications before integration into live experiments.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher – Photonics & Spectroscopy

Jena, Germany

Jun 2020 – Dec 2021

- **Dispersive Spectrometer Design (Applied Sciences '24):** Led optical layout, ZEMAX tolerancing, and construction of a 1064 nm fiber probe Raman spectrometer for clinical tissue applications. Handled detector alignment, wavelength calibration, and fiber coupling from prototype to published instrument.
- **Hyperspectral ML Classification:** Applied supervised machine learning to 2D hyperspectral Raman tissue images, achieving high-accuracy cancer vs. healthy cell separation. Accelerated experimental characterization loops significantly.

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Industrial Vision Pipelines:** Deployed Halcon-based inspection systems on high-speed glass-bottle production lines. Engineered custom bright-field, dark-field, and ring-light illumination schemes; resolved label-verification and liquid-level failures. System reached full customer sign-off.
- **Hardware Specification:** Selected industrial cameras, lenses, and sensors; implemented and validated full optical inspection setups on live production lines.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Precision Optical Metrology

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Competitively selected for ZEISS Summer School (precision optics engineering, Oberkochen). Awarded COLFUTURO Grant 2026 for top Colombian professionals.
- Juan David Munoz Bolanos et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025). Integrated adaptive optics into a confocal Raman system; experimentally resolved and corrected phase aberrations for diffraction-limited deep-tissue imaging.
- Juan David Munoz Bolanos et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024). Full optical design, tolerancing, construction, and clinical validation of a custom dispersive fiber probe spectrometer.
- Sohmen M., Juan David Munoz Bolanos et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023). Integrated optofluidic tuneable lenses into a multiphoton microscope with custom Python/LabVIEW control for fast, aberration-free focusing.

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft

Group leader for Spectroscopy Imaging

christoph.kraft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

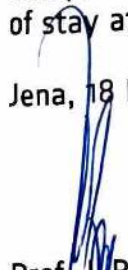
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

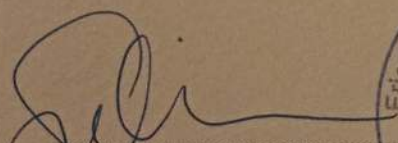
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
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OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIAN

Sexo / Sex Lugar de nacimiento / Place of birth

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

G. ANTIOQUIA

Forma del bulbo / *bulbosus* & *bulbosus*

Dear Family

[illegible]

